

Enterprise Telecom Operations Management Platform

Built on ServiceNow

GRADUATION PROJECT 2026

Information Technology Institute (ITI)
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Chapter 1: Introduction

1.1 Project Overview

The Enterprise Telecom Operations Management Platform is a fully scoped ServiceNow application designed to automate and streamline telecom customer complaint handling, network outage management, and field engineer dispatching. The system was developed as part of the ITI Graduation Project 2026, leveraging the ServiceNow Personal Developer Instance (PDI) under the application scope x_telcom_ops.

Property	Details
Project Name	Enterprise Telecom Operations Management Platform
Platform	ServiceNow PDI (Personal Developer Instance)
Application Scope	x_telcom_ops
Application Type	Scoped Application
Project Duration	8-9 Hours (11 Phases)
Project Level	ServiceNow Fundamentals + Development
University	Information Technology Institute (ITI)
Year	2026

1.2 Problem Statement

The telecom industry faces significant operational challenges that directly impact customer satisfaction and business efficiency:

- Customers experience long resolution times when reporting network complaints such as no signal, slow internet, call drops, and billing issues.
- There is no unified system to track complaint lifecycle from submission through resolution.
- Network outages are managed manually with no automated escalation to field engineering teams.
- Agents lack visibility into related incidents, outages, and engineer tasks from a single view.
- There is no automated compensation mechanism for customers affected by service outages.
- Reporting and KPI tracking for operations managers is limited and fragmented.

1.3 Project Objectives

The system aims to achieve the following objectives:

- Enable customers to self-register and access services through a dedicated Service Portal.
- Allow customers to raise telecom complaints (No Network, Slow Internet, Call Drop, Billing Issue, SIM Issue) directly from the portal.
- Enable complaint agents to manage, track, and resolve customer complaints through structured workflows.

- Automatically generate ITSM Incident records upon complaint creation, linking them to the source complaint.
- Allow network operations teams to create and manage Network Outage records linked to complaints.
- Automatically dispatch Field Engineer Tasks when a network outage is recorded.
- Automatically close outages when engineer tasks are marked as Work Completed.
- Provide SLA enforcement for complaints and outages with automated breach notifications.
- Provide a reporting dashboard for operations managers, complaint agents, and network engineers.
- Automate email notifications across all key events in the workflow.

1.4 Scope & Limitations

Customer Scope

- Register as a customer through the Service Portal.
- Raise complaints and track their status.
- View active network outages in their area.
- Submit compensation requests via the portal.
- Receive automated email notifications on complaint progress.

Admin / Agent Scope

- Manage customer records, service plans, and complaints from the back-end platform.
- Access auto-created Incident and Engineer Task records.
- Configure SLAs, notifications, and ACL security rules.
- View operations dashboard and generate reports.

Limitations

- The system was built on a free ServiceNow PDI (Personal Developer Instance); no production-grade infrastructure was used.
- Advanced AI-based prediction features are out of scope for this version.
- Integration with live telecom billing systems is not included.

Chapter 2: Literature Review / Background

2.1 ServiceNow Platform Overview

ServiceNow is a cloud-based platform-as-a-service (PaaS) solution that provides IT Service Management (ITSM), Customer Service Management (CSM), and many other enterprise workflow automation capabilities. Originally designed as an ITSM tool, it has evolved into a broad enterprise platform used across industries including telecommunications, healthcare, finance, and manufacturing.

Key platform capabilities include: a table-based data model, scripting with JavaScript (GlideScript), Flow Designer for no-code automation, Service Portal for end-user experiences, and robust reporting and dashboarding.

2.2 ITSM – IT Service Management

ITSM (IT Service Management) on ServiceNow provides structured processes for managing IT services. The Incident module, which is a core part of ITSM, was used in this project to automatically track technical issues arising from customer complaints. The Incident table (incident) is a native ServiceNow table that extends the Task table, providing fields such as State, Priority, Assigned To, Assignment Group, and Activity journal.

2.3 CSM – Customer Service Management

CSM (Customer Service Management) focuses on managing end-customer interactions. While this project implements a custom CSM-like solution within a scoped application, the principles of CSM are applied: customer registration, complaint lifecycle management, and automated agent assignment.

2.4 Field Service Management (FSM)

Field Service Management involves dispatching field workers (engineers) to resolve on-site technical problems. In this project, a custom Field Engineer Task table (x_telcom_field_engineer_task) was created to model engineer dispatch workflows, including status tracking (Assigned, Traveling, On Site, Work Completed) and automatic outage closure upon completion.

2.5 Why ServiceNow for Telecom?

ServiceNow is widely adopted in the telecom industry for several reasons:

- Unified platform: Complaints, incidents, outages, and engineer tasks can all be managed from a single system.
- Automation: Business Rules, Flow Designer, and SLA policies automate repetitive tasks, reducing human error.
- Self-service portal: Customers can interact with the company without requiring agent intervention.
- Scalability: The platform scales from a single-department tool to an enterprise-wide system.
- Reporting and KPIs: Built-in reporting allows operations managers to monitor performance in real time.

- Security: Role-based ACL (Access Control Lists) ensure data privacy and appropriate data visibility.
- Integration: ServiceNow integrates with external systems via REST APIs, enabling ecosystem connectivity.

Chapter 3: Requirements

3.1 Functional Requirements

#	Requirement	Actor
FR-01	Customer can register via Service Portal	Customer
FR-02	Customer can raise a telecom complaint from the portal	Customer
FR-03	System automatically creates an Incident when a complaint is submitted	System
FR-04	Complaint agent can manage, update, and resolve complaints	Complaint Agent
FR-05	System automatically assigns complaint to the Complaint Resolution Team	System
FR-06	Network team can create and manage Network Outage records	Network Ops Team
FR-07	System automatically creates a Field Engineer Task when an outage is recorded	System
FR-08	Outage automatically closes when the Engineer Task status = Work Completed	System
FR-09	Customer can submit a compensation request via the portal	Customer
FR-10	Operations Manager can approve or reject compensation requests	Operations Manager
FR-11	System sends automated email notifications at every key event	System
FR-12	SLA timers are enforced on complaints and outages based on priority/severity	System
FR-13	Reports and a dashboard are available for the Operations Manager	Operations Manager
FR-14	Customer can only see their own complaints (ACL-enforced)	System / ACL

3.2 Non-Functional Requirements

Category	Requirement
Security	Role-based ACL rules ensure customers can only view their own records. Separate groups control access to complaint, outage, engineer, and compensation tables.
Performance	SLA definitions enforce time-based resolution targets. Flows and business rules

	execute automatically without manual intervention.
Usability	The Service Portal provides a clean, self-service experience. Forms use UI Policies and Client Scripts to guide users dynamically.
Reliability	Business Rules and Flow Designer provide redundant automation paths for critical operations such as Incident creation and Engineer Task dispatch.
Maintainability	All custom objects are scoped under x_telcom_ops, isolating the application from global platform configurations.
Scalability	The data model is extensible. New complaint types, service plans, or engineer roles can be added without breaking existing workflows.

Chapter 4: System Design

4.1 Application Scope

The application was developed as a Scoped Application in ServiceNow under the scope prefix `x_telcom_ops`. All custom tables, scripts, flows, portals, and security rules are namespaced within this scope to ensure isolation from the global platform and other applications.

4.2 Data Model

The application uses six custom tables. Three of them extend the native Task table to inherit workflow fields (State, Priority, Assigned To, Assignment Group, Comments, Work Notes, Due Date). The remaining three are standalone custom tables.

#	Table Name	System Name	Extends
1	Customer Profile	x_telcom_customer_profile	None (custom)
2	Service Plan	x_telcom_service_plan	None (custom)
3	Telecom Complaint	x_telcom_customer_complaint	Task Table
4	Network Outage	x_telcom_network_outage	Task Table
5	Field Engineer Task	x_telcom_field_engineer_task	Task Table
6	Customer Compensation	x_telcom_customer_compensation	None (custom)

Why Extend the Task Table?

The Task table (task) is the foundational ServiceNow table that provides standard workflow fields used across ITSM processes. Extending it for Complaint, Outage, and Engineer Task tables gives those tables automatic access to:

- Number (auto-generated, renamed per table)
- State (lifecycle management)
- Priority and Urgency
- Assigned To and Assignment Group (team routing)
- Comments (customer-visible journal)
- Work Notes (internal-only journal)
- Due Date and SLA integration

Table 1 — Customer Profile (x_telcom_customer_profile)

Stores telecom customer details. Does not extend Task.

Field Label	Field Type	Choice Values / Reference
Customer Number	Auto Number	—
Full Name	String	—

Mobile Number	Phone Number	—
Email Address	Email	—
Address	String	—
City	String	—
State	Choice	NY, CA, TX, FL, IL
ZIP Code	String	—
Service Plan	Reference	x_telcom_service_plan
Customer Type	Choice	Prepaid, Postpaid, Enterprise
Customer Status	Choice	Active, Suspended, Disconnected
User Account	Reference	sys_user

Table 2 — Service Plan (x_telcom_service_plan)

Stores available telecom service plans.

Field Label	Field Type	Choice Values
Plan Number	Auto Number	—
Plan Name	String	—
Plan Type	Choice	Mobile, Broadband, Fiber
Monthly Cost	Currency	—
Data Limit	Integer	—
Call Limit	Integer	—
SMS Limit	Integer	—
Plan Status	Choice	Active, Retired

Table 3 — Telecom Complaint (x_telcom_customer_complaint) — Extends Task

Manages customer complaint records. The auto-generated Number field is renamed to Complaint Number.

Inherited Task fields used: State (New, Open, Work In Progress, Pending, Resolved, Closed), Priority (1-Critical to 4-Low), Assigned To, Assignment Group, Comments, Work Notes.

Field Label	Field Type	Choice Values / Reference
Customer	Reference	x_telcom_customer_profile
Complaint Type	Choice	No Network, Slow Internet, Call Drop, Billing Issue, SIM Issue
Complaint Channel	Choice	Portal, Call Center, Store

Impact Level	Choice	Critical, High, Medium, Low
Area Impacted	String	—
Resolution Status	Choice	Open, In Progress, Pending Customer, Resolved, Closed
Related Outage	Reference	x_telcom_network_outage
Related Incident	Reference	incident

Table 4 — Network Outage (x_telcom_network_outage) — Extends Task

Tracks network outage events. The Number field is renamed to Outage Number.

Field Label	Field Type	Choice Values / Reference
Outage Type	Choice	Tower Down, Fiber Cut, Power Failure, Network Congestion
Impacted Area	String	—
Impacted Customers	Integer	—
Severity	Choice	Critical, High, Medium, Low
Root Cause	HTML	—
Outage Status	Choice	Open, Investigating, Engineer Assigned, Resolved, Closed
Related Incident	Reference	incident

Table 5 — Field Engineer Task (x_telcom_field_engineer_task) — Extends Task

Tracks engineer repair work. The Number field is renamed to Engineer Task Number.

Field Label	Field Type	Choice Values / Reference
Related Outage	Reference	x_telcom_network_outage
Engineer	Reference	sys_user
Engineer Status	Choice	Assigned, Traveling, On Site, Work Completed
Estimated Arrival	Date/Time	—
Resolution Notes	HTML	—

Table 6 — Customer Compensation (x_telcom_customer_compensation)

Stores customer compensation records. Does not extend Task.

Field Label	Field Type	Choice Values / Reference
Compensation Number	Auto Number	—

Customer	Reference	x_telcom_customer_profile
Related Complaint	Reference	x_telcom_customer_complaint
Compensation Type	Choice	Bill Credit, Refund, Free Data
Compensation Amount	Currency	—
Approval Status	Choice	Pending, Approved, Rejected

4.3 Table Relationships Diagram

The following describes the data flow between tables in the application:

- Customer → Complaint: A customer can raise multiple complaints. The Complaint table references Customer Profile.
- Complaint → Incident: Each complaint automatically creates one Incident record (native incident table).
- Complaint → Network Outage: Complaints can be linked to a related network outage.
- Network Outage → Field Engineer Task: Each outage automatically generates one or more engineer tasks.
- Customer → Compensation: Compensation records are linked back to the customer and the original complaint.
- Task Table (parent): Complaint, Outage, and Engineer Task all extend the Task table, inheriting core workflow fields.

4.4 Roles & Groups

Role / Group	Tables Accessible	Permissions
Customer (portal role)	Customer Profile, Complaint, Compensation	Read/Write own records only (ACL enforced)
Complaint Resolution Team	Complaint, Incident	Full read/write on complaints
Network Operations Team	Network Outage, Complaint (read)	Full read/write on outages
Field Engineering Team	Field Engineer Task, Outage (read)	Full read/write on engineer tasks
Operations Manager	All tables, Dashboard, Reports	Full access including compensation approval
ITSM Support Team	Incident	Standard ITSM incident management

Chapter 5: Implementation

5.1 Phase 1 — Application Setup

In this phase, the ServiceNow Personal Developer Instance (PDI) was provisioned and a new scoped application was created with the scope `x_telcom_ops`. The application was named 'Enterprise Telecom Operations Management Platform'. All subsequent development was performed within this scoped context to ensure isolation from the global platform.

5.2 Phase 2 — Data Model (Tables)

Six custom tables were created as described in Chapter 4. The three tables that extend the Task table (Complaint, Network Outage, Field Engineer Task) automatically inherited all Task fields. The Number field on each Task-extending table was renamed to match the table's domain (Complaint Number, Outage Number, Engineer Task Number). Related Lists were configured as follows:

Parent Table	Related List Added
Customer	Complaints
Customer	Compensation
Complaint	Outages
Outage	Engineer Tasks

5.3 Phase 3 — Forms & UI

Form layouts were configured for each table to present the most important fields in logical sections. Tab-based layouts were applied to complex forms:

Complaint form sections: Customer Information | Complaint Details | Escalation Information | Activity

Outage form sections: Outage Information | Work Information | Activity

Engineer Task form sections: Engineer Information | Work Information | Activity

5.4 Phase 4 — Security Configuration (ACL)

Five ACL rules were created to control data access:

ACL	Table	Access Rule
ACL 1	Customer Table	Customers can see only their own records (condition: <code>gs.getUserID() == current.user_account</code>)
ACL 2	Complaint Table	Only the Complaint Resolution Team group has access
ACL 3	Network Outage Table	Only the Network Operations Team group has access
ACL 4	Engineer Task Table	Only the Field Engineering Team group has access
ACL 5	Compensation Table	Only the Operations Manager role has access

5.5 Phase 5 — Client-Side Scripting

Three client scripts were created to provide real-time form validation and dynamic field updates:

Client Script 1 — Mobile Number Validation (onSubmit)

Table: Customer Profile. Logic: If mobile number length is less than 10 digits, display an error message and stop form submission.

Client Script 2 — Auto Priority (onChange)

Table: Complaint. Logic: When Impact Level changes, automatically set Priority. Critical → 1, High → 2, Medium → 3, Low → 4.

Client Script 3 — Engineer Validation (onSubmit)

Table: Engineer Task. Logic: If the Engineer field is empty upon form submission, display an error and prevent saving.

5.6 Phase 6 — Server-Side Scripting

UI Policies

UI Policy	Table	Condition	Action
UI Policy 1 — Critical Complaint	Complaint	Impact Level = Critical	Show and make Escalation Comments field mandatory
UI Policy 2 — Outage Required	Complaint	Complaint Type = No Network	Make Related Outage field mandatory

Data Policies

Data Policy	Table	Field	Behavior
Data Policy 1 — Mobile Mandatory	Customer	Mobile Number	Mandatory = True (enforced server-side)
Data Policy 2 — Description Required	Complaint	Description	Mandatory = True

Business Rules

BR	Table	When	Purpose
BR 1 — Create Portal User	Customer	After Insert	Auto-create sys_user, copy name/email, assign customer role
BR 2 — Prevent Duplicate	Customer	Before Insert	Check for duplicate mobile number, abort if found
BR 3 — Auto Assign Complaint	Complaint	Before Insert	Assign Complaint Resolution Team automatically

BR 4 — Create Incident	Complaint	After Insert	Create Incident record, copy details, link to complaint
BR 5 — Create Engineer Task	Outage	After Insert	Auto-create Field Engineer Task linked to outage
BR 6 — Close Outage	Engineer Task	After Update	When Engineer Status = Work Completed, close the related Outage

Script Includes & GlideAjax

Script Include	Purpose	Functions
CustomerUtils	Customer validation utilities	validateCustomer(), checkDuplicateCustomer()
ComplaintUtils	Complaint processing utilities	setPriority(), assignmentLogic()
EngineerUtils	Engineer handling utilities	createEngineerTask(), findEngineer()

5.7 Phase 7 — Automation (Flow Designer)

Six flows were created in Flow Designer to automate the end-to-end complaint resolution lifecycle:

Flow	Trigger	Actions
Flow 1 — Customer Registration	Customer record created	Create sys_user → Assign role → Send welcome notification
Flow 2 — Complaint Flow	Complaint record created	Set priority → Assign complaint group → Create incident → Send notification
Flow 3 — Incident Flow	Incident record created	Assign ITSM Support Team → Notify support team
Flow 4 — Outage Flow	Outage record created	Assign Network Ops Team → Create engineer task → Send outage notification
Flow 5 — Engineer Flow	Engineer task record updated	Update outage state → Close outage (if Work Completed)
Flow 6 — Compensation Flow	Compensation record created	Send approval → Notify customer

5.8 Phase 8 — SLA & Notifications

SLA Definitions

Complaint SLA — Start: State = Open | Pause: State = Pending | Stop: State = Resolved OR Closed

Priority	Resolution Time
Critical	1 Hour

High	4 Hours
Medium	8 Hours
Low	24 Hours

Outage SLA — Start: State = Open | Stop: State = Resolved

Severity	Resolution Time
Critical	30 Minutes
High	2 Hours
Medium	4 Hours
Low	8 Hours

Events & Email Notifications

Event Name	Notification	Trigger
customer.created	Welcome Notification	Customer registration completed
complaint.created	Complaint Confirmation	Complaint submitted by customer
incident.created	Incident Created Alert	Incident auto-created from complaint
outage.created	Outage Alert	New network outage recorded
engineer.assigned	Engineer Assigned	Field engineer dispatched to site
compensation.approved	Compensation Approved	Compensation request approved by manager

5.9 Phase 9 — Service Portal

The Telecom Customer Portal was built using the ServiceNow Service Portal framework. The portal uses Record Producers, Data Table widgets, and Simple List widgets. No advanced Angular custom widgets were created (keeping within beginner scope).

Page	Component Type	Key Fields / Columns
1. Home Page	Welcome + Quick Links widgets	Portal intro, navigation links, My Complaints widget, Active Outages widget
2. Customer Registration	Record Producer	Full Name, Mobile Number, Email, Address, Service Plan
3. Raise Complaint	Record Producer	Complaint Type, Area Impacted, Impact Level, Description
4. My Complaints	Data Table Widget	Number, State, Priority, Assigned To, Created
5. Outage Status	Simple List Widget	Number, Area, Severity, State

6. Compensation Request	Record Producer	Complaint, Compensation Type, Reason
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5.10 Phase 10 — Reports & Dashboard

Dashboard: Telecom Operations Dashboard — used by Operations Manager, Complaint Agents, and Network Engineers.

Dashboard Widgets

Widget	Type	Table	Purpose
1. Open Complaints	KPI	Complaint	Count active complaints (State not Closed)
2. Active Outages	KPI	Outage	Count active outages (State not Resolved)
3. SLA Breaches	Bar Chart	Task SLA	Track SLA violations by type
4. Engineer Workload	Pie Chart	Engineer Task	Workload per engineer (Group By Assigned To)
5. Complaint Trend	Line Chart	Complaint	Complaint volume over time (Group By Created Date)

Standalone Reports

Report	Type	Group By	Purpose
Complaints by Type	Pie Chart	Complaint Type	Show most common complaint categories
Open Outages	List Report	—	View all currently active outages
Engineer Task Status	Bar Chart	State	Track engineer task progress by status
SLA Breach Report	List Report	—	Identify all breached SLA records
Complaint Trend Report	Line Chart	—	Track complaint volume increase/decrease over time

5.11 Phase 11 — Final Demo Flow

The complete end-to-end demonstration flow follows these 12 steps:

- 1. Customer Registration — Customer registers via Service Portal Record Producer
- 2. Portal Login — Customer logs into the portal using auto-created credentials
- 3. Raise Complaint — Customer submits a complaint (e.g., No Network in Area X)
- 4. Complaint Assignment — System auto-assigns to Complaint Resolution Team
- 5. Incident Creation — Incident automatically created and linked to complaint
- 6. Outage Creation — Network team creates a Network Outage record
- 7. Engineer Task Creation — Engineer Task auto-created and assigned to field engineer

- 8. Engineer Resolution — Engineer updates status to Work Completed
- 9. Outage Closure — Outage automatically closes when engineer task is completed
- 10. SLA Tracking — SLA timers monitored throughout; breach alerts triggered if exceeded
- 11. Notification Trigger — Email notifications sent at each key event
- 12. Dashboard Update — Operations Manager views updated KPIs and charts in real time

Chapter 6: Testing

6.1 Test Cases

TC#	Test Case	Expected Result	Status
TC-01	Customer registers via portal with valid data	Customer record created; sys_user created; welcome email sent	[Screenshot]
TC-02	Customer registers with duplicate mobile number	System prevents duplicate; error message shown	[Screenshot]
TC-03	Customer raises complaint with Impact Level = Critical	Priority auto-set to 1-Critical; Escalation Comments field shown	[Screenshot]
TC-04	Complaint submitted with Complaint Type = No Network	Related Outage field becomes mandatory	[Screenshot]
TC-05	Complaint submitted and saved	Incident automatically created and linked to complaint	[Screenshot]
TC-06	Network Outage record created	Field Engineer Task auto-created; Network Ops Team assigned	[Screenshot]
TC-07	Engineer updates Engineer Status = Work Completed	Related Outage automatically closed	[Screenshot]
TC-08	Complaint SLA — Critical priority; no resolution in 1 hour	SLA breach notification triggered	[Screenshot]
TC-09	Customer views My Complaints page in portal	Only the logged-in customer's complaints are visible (ACL test)	[Screenshot]
TC-10	Compensation request submitted and approved	Customer receives compensation approved email notification	[Screenshot]

6.2 Test Scenarios — End-to-End Flow

The following end-to-end scenario was tested to validate the complete system:

Customer Registration → Portal Login → Raise Complaint (No Network, Critical) → Auto-assignment to Complaint Team → Auto-creation of Incident → Outage Creation by Network Team → Auto-creation of Engineer Task → Engineer marks Work Completed → Outage closes → Customer submits Compensation → Manager approves → Email sent to customer

Each step was verified by checking the related records in ServiceNow, confirming the automated links, and validating that email notifications were sent correctly.

Chapter 7: Results & Discussion

7.1 What Works

- Customer self-registration via Service Portal with automatic sys_user creation.
- Complaint submission with dynamic form behavior (auto-priority from impact level, mandatory fields via UI Policies).
- Automatic Incident creation upon complaint submission with full field linking.
- Automatic Field Engineer Task creation upon outage recording.
- Automatic outage closure when engineer marks work as completed.
- SLA timers enforcing resolution deadlines for complaints and outages.
- Email notifications at all six key events in the workflow.
- ACL enforcement ensuring customers see only their own records.
- Operations dashboard providing real-time KPIs for management.
- Full demo flow validated from customer registration to compensation approval.

7.2 Challenges Faced & How Solved

Challenge	How Solved
ACL rule preventing agents from seeing complaint records they should access	Reviewed ACL role conditions; added correct group membership checks using <code>gs.getUser().isMemberOf()</code>
Business Rule for auto-creating Incident was running in global scope instead of <code>x_telcom_ops</code>	Ensured the Business Rule was created inside the scoped application context
Portal Record Producer not linking the customer reference correctly	Added a script variable to capture the logged-in user's customer profile record and map it to the Customer field
SLA not starting because State field choice values did not match the SLA condition string	Corrected the State choice values to match exactly with SLA start/stop conditions
Flow Designer not triggering on record insert	Changed the trigger from 'Updated' to 'Created' and re-activated the flow after testing

7.3 Key Learnings

During this project, the following ServiceNow concepts were applied and understood:

- ACL (Access Control Lists) — Role and condition-based data visibility
- Flow Designer — No-code/low-code automation with trigger → action patterns
- Business Rules — Server-side scripting for insert/update automation
- Client Scripts — `onLoad`, `onChange`, `onSubmit` form behaviors
- UI Policies — Dynamic field visibility and mandatory enforcement
- Data Policies — Server-side mandatory field enforcement
- Script Includes — Reusable server-side utility classes
- GlideAjax — Asynchronous client-server communication

- SLA (Service Level Agreements) — Time-based compliance enforcement
- Events & Notifications — Event registry and email notification configuration
- Service Portal — Record Producers, Data Table, Simple List widgets
- Table Relationships — Reference fields and schema map
- Task Table Extension — Inheritance of workflow fields
- Scoped Applications — Namespacing and application isolation
- Reports & Dashboards — KPI, bar, pie, and line chart creation

Chapter 8: Conclusion & Future Work

8.1 Summary

The Enterprise Telecom Operations Management Platform successfully demonstrates how ServiceNow can be used to automate and streamline telecom customer service operations. The project was implemented across 11 structured phases, covering the full ServiceNow development lifecycle from application setup and data modeling through to security, automation, portal development, and reporting.

The team successfully delivered all planned features: a self-service customer portal, automated complaint-to-incident-to-engineer workflow, SLA enforcement, email notification system, and an operations dashboard. The project demonstrates practical competence in ServiceNow fundamentals relevant to the CSA certification and beyond.

8.2 Future Enhancements

- **AI Integration:** Use ServiceNow's Now Intelligence (AI/ML) to automatically predict complaint resolution time and priority based on historical data.
- **Virtual Agent (Chatbot):** Implement a ServiceNow Virtual Agent chatbot on the Service Portal to guide customers through complaint submission without manual navigation.
- **Predictive Outage Detection:** Integrate with network monitoring APIs to proactively detect and create outage records before customers report them.
- **Mobile App:** Extend the portal experience to a native mobile application for customer complaint submission.
- **Live Tracking:** Allow customers to track the real-time location and ETA of dispatched field engineers.
- **Billing System Integration:** Integrate with external telecom billing APIs to automate compensation credit application directly to customer accounts.
- **Multi-language Support:** Add Arabic language localization to the Service Portal for broader user accessibility.
- **Advanced Analytics:** Implement Performance Analytics dashboards for trend forecasting and SLA risk prediction.

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- ITSM Best Practices — ITIL v4 Foundation Guide
- ServiceNow Community — <https://community.servicenow.com>

Appendices

Appendix A — Application Scope Details

Application Name: Enterprise Telecom Operations Management Platform

Scope Prefix: x_telcom_ops

All tables, scripts, flows, and portal configurations are prefixed with x_telcom_ops to maintain application isolation.

Appendix B — Important Code Snippets

Business Rule 4 — Create Incident from Complaint (After Insert)

```
(function executeRule(current, previous) {  
    var inc = new GlideRecord('incident');  
    inc.initialize();  
    inc.short_description = 'Incident from Complaint: ' + current.number;  
    inc.description = current.description;  
    inc.priority = current.priority;  
    var incSysId = inc.insert();  
    current.x_telcom_ops_related_incident = incSysId;  
    current.update();  
})(current, previous);
```

Business Rule 6 — Close Outage (After Update on Engineer Task)

```
(function executeRule(current, previous) {  
    if (current.x_telcom_ops_engineer_status == 'work_completed') {  
        var outage = current.x_telcom_ops_related_outage.getRefRecord();  
        outage.x_telcom_ops_outage_status = 'Closed';  
        outage.update();  
    }  
})(current, previous);
```

Client Script 2 — Auto Priority Based on Impact Level (onChange)

```
function onChange(control, oldValue, newValue, isLoading) {  
    if (isLoading || newValue === '') return;  
    var priorityMap = {  
        'Critical': '1', 'High': '2', 'Medium': '3', 'Low': '4'  
    };  
    g_form.setValue('priority', priorityMap[newValue] || '4');  
}
```


Appendix C — Screenshot Placeholders

The following screenshots should be inserted in the final submitted document:

- [Screenshot 1] — Application Scope setup in ServiceNow Studio
- [Screenshot 2] — All six custom tables in the Schema Map
- [Screenshot 3] — Customer Profile form layout
- [Screenshot 4] — Complaint form with all sections
- [Screenshot 5] — ACL rules for all five tables
- [Screenshot 6] — Client Scripts list in the application
- [Screenshot 7] — Business Rules list in the application
- [Screenshot 8] — Flow Designer — Complaint Flow steps
- [Screenshot 9] — SLA definitions for Complaint and Outage tables
- [Screenshot 10] — Email notification templates
- [Screenshot 11] — Service Portal — Home Page
- [Screenshot 12] — Service Portal — Raise Complaint Record Producer
- [Screenshot 13] — Service Portal — My Complaints page
- [Screenshot 14] — Telecom Operations Dashboard
- [Screenshot 15] — End-to-end demo: complaint → incident → engineer task → outage closed